

The Blind Watchmaker

Harmonic Latent Space and the Geometry of Bias and Variance

Ryan J. Brooks

Abstract

Do you exist in other people’s realities the same way you exist in your own? It is a question the field tends to defer: the reflex is to hand the burden of proof back to whoever raised it, as if demanding evidence were the neutral posture and asking was the indulgence. That reflex is not neutral. It is a prior, learned so early in life it passes for common sense. The dominant program in machine learning rests on one such prior: that a single architecture, given enough parameters, data, and compute, can reach anything worth learning - that the architecture itself is neutral ground. This is the Universal Approximation Theorem, and we do not contest it. The theorem is a promise of existence: with enough width, a network can represent the target. It is silent on what follows - which of those representable functions finite training actually reaches, and at what cost. Our claim lives in that silence, alongside the theorem rather than against it: what a model learns in finite training is set less by how large it is than by the geometry of its representation - the shape of the space its weights are allowed to occupy. Approximation says the destination exists; geometry says which destinations are reachable, and how cheaply. We make that concrete with two components of the Praxis framework [7]: a harmonic head that writes a learned standing wave over its features, and the CALM continuous-latent codec [34], which phase-locks a compressed patch latent to that rhythm. The same geometry reaches its finest, per-feature form in a plain transformer, so across these approaches it is the granularity that changes, not the geometry. In both, the classical bias-variance trade-off stops being a single dial. A static, population-average rhythm (the spectrum the field phase-locks to) is pure bias; an input-conditional modulation of that rhythm is structured variance; and because the two live on orthogonal axes of a harmonic latent space, they can be lowered together rather than traded against each other. This reframing motivates a central conjecture: consensus modeling - learning the mean of a corpus - scales linearly in the patterns it must express, while harmonic modeling, which composes patterns by interference, scales logarithmically, because K frequency components address exponentially many configurations. The framework exists so that the reader can test it.

1 Introduction

The dominant recipe for capability is scale. Fix a transformer [43], then grow it: more parameters, more tokens, more compute, with returns described by smooth scaling laws [22]. The recipe works, but it assumes the architecture itself imposes no ceiling on which patterns gradient descent can find - that everything worth learning is reachable from any sufficiently large model. This paper questions that assumption and offers a concrete alternative: that the *geometry* a model is allowed to represent, not its size, determines the patterns accessible to it during finite training.

With geometry as defined (Section 2), the objects in play are *symbols* - discrete geometries the model reuses - and what it learns is a change in symbolism over time. The oldest tradeoff in statistics gives the first coordinate on that change. A model that minimizes error by collapsing toward the average of its training data has high bias and low variance; a model that chases the particulars of each example has low bias and high variance. Classically these trade off along a single line: you choose a point on it. Our first contribution is to show that in a *continuous, harmonic* latent space this is no longer true. Bias and variance become separate coordinates, and a model can move in one without paying in the other.

We ground this in mechanisms that already run in Praxis, not in metaphor. The harmonic head (Section 3) builds a two-dimensional standing wave over (position, feature) from a learned grid of frequency amplitudes with irrational, frozen phases. A *static* amplitude grid produces the same field for every input: it is the corpus-average rhythm, maximum bias, a flat-by-construction prior. Making those amplitudes depend on the input adds *structured* variance - deviation that still respects the learned harmonic constraints. The CALM codec (Section 4) shows the same physics from the other side: a continuous-latent autoregressor that reconstructs sequences near-perfectly by phase-locking its latent features to a dominant rhythm. From these two we draw a scaling conjecture (Section 5): the mean costs linearly, interference costs logarithmically.

This paper therefore contributes: (1) a reframing of bias and variance as orthogonal axes of a harmonic latent space, grounded in a running implementation and its measured diagnostics; (2) a conjecture that consensus modeling scales linearly while harmonic modeling scales logarithmically, stated honestly as a conjecture; (3) the Praxis framework for testing it; and (4) a candid account of what is measured, what is conjectured, and what remains philosophy.

2 Definitions

The paper leans on a small vocabulary, used precisely. It is ordinary signal processing; the only commitment is to read the model’s representations through it.

Harmonic. A representation is *harmonic* when it is carried by a superposition of oscillatory modes over a fixed frequency basis - amplitudes and phases on a frequency lattice, inverse-transformed into the signal - rather than by values in the raw signal space. The field $b = \text{IRFFT}(a e^{i\varphi}/f^\alpha)$ of Section 3 is the concrete instance: a is a spectrum of amplitudes, φ a set of phases, f the frequency lattice. The word is not shorthand for “oscillating” or “smooth”; it is the literal claim that the representation is a sum of fixed modes and that only their amplitudes vary.

Amplitude, frequency, phase, velocity. A mode at frequency f contributes $a_f \cos(2\pi f x + \varphi_f)$: *amplitude* a_f is how much of the mode is present, *phase* φ_f is where in its cycle it sits, *frequency* f is which mode it is. A mode’s frequency is also its angular *velocity* - the rate its phase advances - so for a single mode “frequency” and “velocity” name the same quantity, and a velocity that varies across the signal is simply a frequency that changes (a chirp). In Praxis the amplitudes are learned, the frequencies are a fixed lattice, and the phases are *frozen*.

Frozen phases. The phases are seeded once - by Weyl equidistribution (Section 3) - and never trained. This is what fixes the basis: the same modes describe every input, and learning moves only the amplitudes. A representation whose phases are learned per input has no fixed basis and is not harmonic in our sense.

Geometry. The superposition of modes at one moment is a point - a *geometry* - in the latent space; inputs that share an amplitude configuration form a cluster of such points (the crystal-head centers, Figure 8). “Overlaid geometries” are several modes present at once. Distinct modes are orthogonal (the spectral theorem), so independent quantities can occupy separate modes without interacting - the property Section 3 uses to pull bias and variance apart.

Symbol. A *symbol* is a discrete, reusable geometry - a configuration the model commits to and recombines, not a one-off point. The crystal-head centers (Figure 8) are symbols in this sense: a finite, sparsely populated set of geometric units that recur across inputs. A symbol can be a geometry, but not every geometry is a symbol - the added content is that it is discrete and reused. We mean this in the literal, measurable sense (a clustered, low-dimensional set of recurring configurations), not the semiotic one. Read across training, this set is what grokking discovers - the periodic algorithm a network recovers only after its delayed memorize-then-generalize transition [29, 33]; a model with frozen phases is handed that basis at init, so it begins *post-grok* rather than working toward the transition. Reading computation as a *change in symbolism over time* - which symbol is active, and how it shifts across position and depth - is the temporal companion to the static picture above: the symbol set is fixed, and what moves is which symbol is active, re-fitted per input (the *always grokking* conjecture, collected with the others in the Discussion).

Bias and variance, in these terms. *Bias* is the time-invariant part of the

spectrum, the amplitudes shared across all inputs (the corpus rhythm); *variance* is the input-conditional deviation on top, carried by different spectral components and different parameters. The two do not stop being a spectrum: at any single decision boundary, a neuron’s value still sets a *ratio* - how much corpus rhythm and how much input-conditional deviation pass downstream - a point between the ends. What the harmonic latent removes is the assumption that *one* such ratio stands for the whole model. There are many - a consensus of independent decision boundaries, each free to sit at its own point - and it is the aggregate of those ratios, not any single one, that lets bias and variance move as orthogonal coordinates rather than the two ends of a single dial.

Harmonics over time (Koopman). When the representation *evolves* - across sequence position, or across recurrent-depth steps - the harmonic claim is that it evolves approximately *linearly in the mode basis*: each mode carries its own frequency forward and the nonlinear dynamics become a linear operator on the spectrum. This is the Koopman view [25] of a dynamical system, and the sense in which we say the model works “in harmonic space.” The claim has a boundary, and is therefore falsifiable: a representation whose basis depends on the input, that is not a superposition of a fixed set of modes, or whose dynamics are not approximately linear in any fixed basis, is *not* harmonic here. The universality - that *any* dynamics admits such a description - holds only in an infinite-dimensional limit; the working claim is that a *finite* harmonic basis approximates this model’s behavior well enough to measure, and how well is an empirical question, not a definition.

Ghostmax. Attention normalizes scores with *ghostmax* (softmax1, [28]): the denominator is $1 + \sum_j e^{z_j}$ rather than $\sum_j e^{z_j}$, equivalently a single positionless key prepended to every row whose value vector is zero. The extra $e^0 = 1$ is an escape valve - when nothing in the context matches, the head spends its probability mass on the ghost and contributes nothing downstream, instead of being forced to average over irrelevant tokens. It is the dampening valve that lets a head go quiet rather than inject noise the harmonic field would then have to cancel. The boundary: a normalization that must sum to one over the real tokens has no quiet state, and is not ghostmax here.

Dropoff. *Dropoff* is the matched dual of ghostmax. Where the ghost is a zero sink at the *start* of a row that *absorbs* error, dropoff is a zero sink at the *tip* - the most recent position - that *injects* it. In its feature-wise form the value at the causal tip is scaled by an envelope $1 - e^{-(T-1-t)r_d}$ that is zero at the last position and recovers backward at a per-feature rate r_d , so the model cannot lean on the newest token and is pressed to carry its prediction in earlier structure. “Backward” here is the recurrent-depth axis, not sequence time: a causal pass never flows future to past. The boundary, and the present status: ghostmax is in production, but dropoff is a standing hypothesis - wired and runnable, not yet a measured result - and is dropoff only if mass demonstrably migrates off the tip.

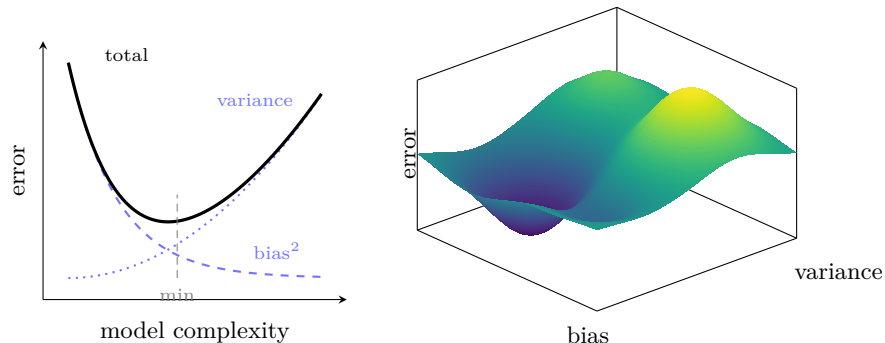


Figure 1: Schematic, not measured. **Left:** the received picture - one knob of effective complexity, so bias^2 falls while variance rises and total error traces a single U with one minimum; the best you can do is find the bottom of that curve. **Right:** when bias and variance are separate coordinates, error becomes a surface over a two-dimensional space - a landscape of basins and ridges navigated in two directions at once, not a single curve to slide along. The decoupling is what turns the line into the landscape.

3 The Mean and the Manifold

3.1 The tradeoff that was supposed to be a line

The bias-variance decomposition treats generalization error as the sum of a term that grows when the model is too rigid to fit the signal (bias) and a term that grows when it is too flexible and fits the noise (variance). The received wisdom is that you cannot reduce both at once with a fixed budget: regularize harder and bias rises; add capacity and variance rises. The tradeoff is drawn as a U-shaped curve with a single minimum, and model selection is the act of finding the bottom of that U.

That picture is correct when the model has one knob mediating both terms - one scalar of effective complexity. It stops being correct when bias and variance are produced by *different parts* of the architecture that can be optimized independently. A harmonic latent space is exactly such an architecture (Figure 1).

3.2 The harmonic field: a measured instance

The prismatic4 head (Praxis, `heads/harmonic.py`) modulates a model’s hidden states multiplicatively by a learned field b :

$$h \mapsto h(1 + b), \quad b[t, d] = \text{IRFFT2}\left(a[f_t, f_d] e^{i\varphi[f_t, f_d]} / f^\alpha\right),$$

where t indexes position, d indexes feature, and $a[f_t, f_d]$ is a learnable grid of amplitudes over a two-dimensional frequency lattice. The phases φ are *not* learned. They are seeded by Weyl’s equidistribution theorem [44]: cell (f_t, f_d)

receives phase $2\pi \text{frac}(f_t\pi + f_d e)$, which is equidistributed on the torus because $\{1, \pi, e\}$ are linearly independent over $\mathbb{Q}$. A radial $1/f^\alpha$ envelope imposes a pink-noise prior. Multiplicative coupling is deliberate: the upstream network cannot cancel the field by emitting $h(x) - b$, because the field scales features rather than adding to them, so learning is forced to flow *through* the field rather than around it.

Now read this through the bias-variance lens.

Bias is the static spectrum. With a fixed amplitude grid, $b[t, d]$ is identical for every input. It is a single rhythm imposed on all of the corpus at once - the population average. In the limit where the grid concentrates onto one cell, every feature oscillates to the same beat and the spectrum heatmap is flat: maximum bias, a model that has collapsed to the mean rhythm of its data. The head exposes this directly. `concentration()` reports Hoyer sparsity of the grid (1 = a single cell, 0 = uniform), and a live spectrum snapshot shows where the energy sits. Early in training the heatmap is indistinguishable from noise; under a smoothness prior it develops a clean low-frequency band - the corpus rhythm crystallizing as a measurable object, not a metaphor. (Figure 2)

Variance is the input-conditional deviation. The decisive architectural step was to stop forcing one static field on a corpus that mixes registers - dialogue, code, prose. A static grid must compromise across all of them, and the compromise degenerates into a content-blind side channel that the rest of the model has to route around. The fix makes the amplitudes a baseline plus an input-conditional delta:

$$a = a_{\text{static}} + \Delta_\phi(\text{context}),$$

where Δ_ϕ is a small network reading multi-scale pooled context, in the spirit of delay embeddings [38], with its output zero-initialized so the field begins as the trained static prior and *anneals* into input dependence. The static baseline carries the bias; the delta carries the variance; an L_2 penalty on the delta keeps the baseline dominant. A diagnostic, `harmonic_delta_norm = rms( $\Delta$ )/rms( $a_{\text{static}}$ )`, reads how much variance the model is choosing to add. Zero means pure bias; a small bounded value means structured variance on top of a stable prior.

The point is that these are *separate parameters with separate gradients*. The smoothness prior controls the bias term; the delta network and its regularizer control the variance term. Lowering one does not mechanically raise the other. The U-shaped curve has become a two-dimensional manifold, and we have coordinates on it.

3.3 Temperature as a coordinate, not as noise

Sampling temperature is usually described as noise injection. On the bias-variance manifold it is better described as a position dial. At low temperature the model commits to the learned rhythm - the high-bias corner. As temperature

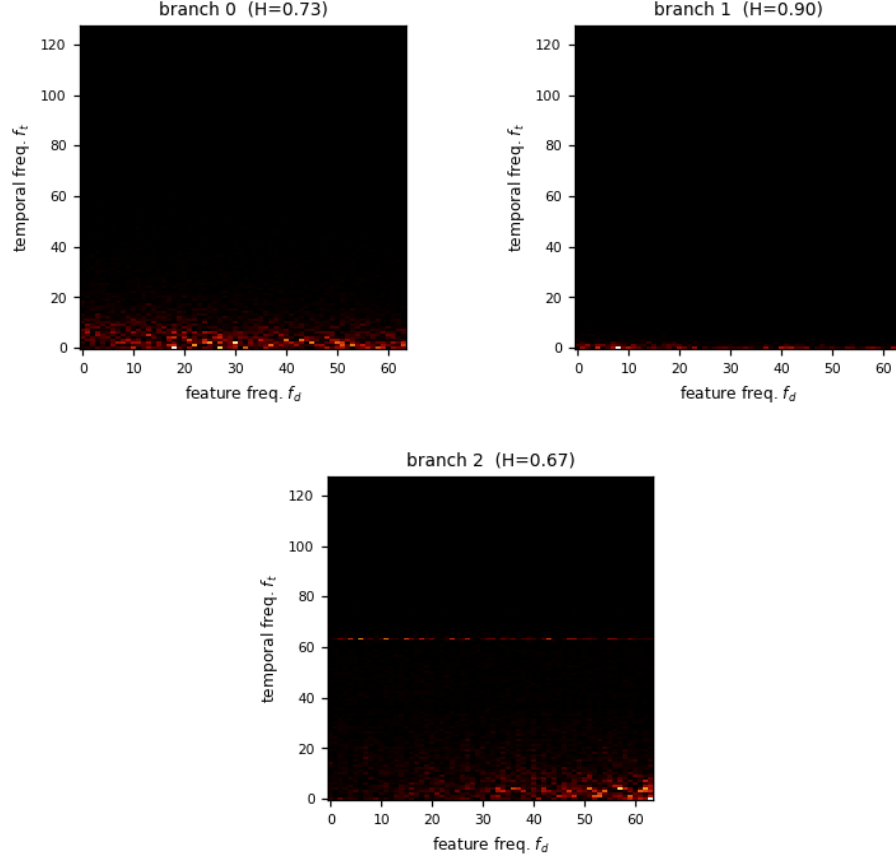


Figure 2: The static amplitude spectrum $|a[f_t, f_d]|$ across the 3 arms of calm-d-1’s prismatic head - the learned grid that, inverse-transformed through the frozen Weyl phases, becomes the bias field. This is the heatmap the text describes: energy that settles into a low-frequency band (small f_t, f_d) is the corpus rhythm crystallizing, while a grid that stays uniform is a field still indistinguishable from noise. Hoyer concentrations $H = 0.73, 0.90, 0.67$ (left to right) quantifies it on the same $[0, 1]$ scale the prose names (1 = a single cell, 0 = uniform), computed from this very tensor by `concentration()` - the same value the dashboard logs live. The pink $1/f^\alpha$ prior lives in the frozen basis, not in these amplitudes, so the grid shown is exactly what the head exposes.

risers it explores within the basin the harmonic constraints define - structured variance. Past a critical point it begins to violate those constraints, which reads as either failure or discovery depending on what you were hoping for. CALM makes this literal: rather than perturbing a latent continuously, it votes over decoded patches [34], so temperature selects how many candidate latents must agree before a patch is emitted. Temperature is not adding randomness to a fixed distribution; it is choosing where on the manifold to stand.

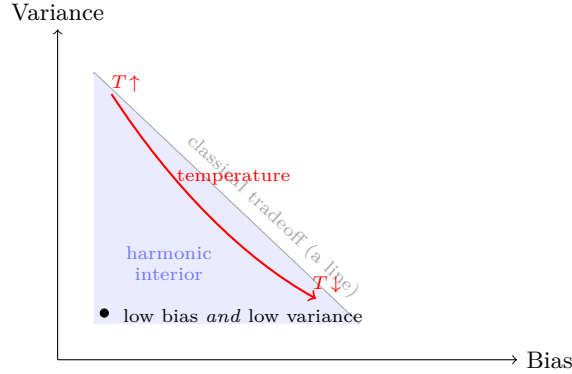


Figure 3: Bias and variance as orthogonal coordinates. Classically a model must sit somewhere on the gray line: less of one means more of the other. When a static spectrum supplies bias and an input-conditional modulation supplies variance, the two decouple, and the interior (shaded) becomes reachable - lower bias *and* lower variance than any point on the line. Temperature is a path through this space, not noise added at a fixed point.

4 Harmonic Latent Space

4.1 Phase-locking and near-perfect reconstruction

CALM trains a variational codec to compress K tokens into a single continuous latent, then models those latents autoregressively with an energy head and decodes by sampling [34]. In Praxis the codec reaches reconstruction in the range of a hundredth of a bit per byte - effectively lossless at the patch level. It does so not by giving each latent feature its own job but by letting the features *phase-lock*: they organize around a dominant rhythm and express the sequence as deviations from it. This is the same phenomenon the harmonic head shows from the bias side. A corpus has a beat; a continuous latent that finds the beat can carry almost all of the signal in a few coherent modes, and spend its remaining capacity on the residual. Reconstruction fidelity is a measure of how completely the latent has phase-locked.

4.2 The seed dissolves

Reproducibility in this regime is no longer an integer. `seed=42` fixes the random number stream that initializes weights and orders data; it does not fix what the trained model *is*. In a harmonic latent space the operative “seed” is the learned field encoded in the weights after training - the frozen phase relationships and the spectrum the amplitudes settled into. Reproducibility becomes a question of which latent dimensions are deterministic (the harmonic structure, fixed by the frozen Weyl phases and the converged spectrum) versus stochastic (the variance envelope, set by temperature and the input-conditional delta). The seed is a property of the trained model, not an argument you pass in. This is uncomfortable for a culture that audits randomness by pinning an integer, and it is the honest situation once the determinism lives in geometry rather than in a stream.

4.3 A memory that is pure bias by construction

The framework also carries a Titans-style neural memory [4] - a small network whose weights are updated online, during the forward pass, by gradient descent on a surprise loss: how badly it reconstructs a value vector from its key. Our variant deviates from the baseline in a way that matters to this paper’s axes. The update is fully detached and layer-wise local, in the Mono-Forward sense [26], itself a descendant of Forward-Forward layer-local learning [20]: no gradient from the task ever reaches the memory’s own learning rule, which is a fixed, scale-invariant step on local surprise. The module is also fully continuous, end to end. It never sees a token, a position, or a logit - continuous vectors arrive, shifted continuous targets leave - so it cannot know what its inputs mean or what its outputs will be used for; in that respect it shares a direction with fully-differentiable computation efforts such as neurallambda [14], which push symbolic machinery into the continuous regime rather than the reverse. This is a zero-knowledge discipline [16] turned inward: the module does verifiable local work - it provably reduces its own surprise - on a witness whose semantics are structurally withheld from it, the same expose-the-output, hide-the-perturbable-witness principle that makes the rest of the system robust to its own observers.

What that surprise targets is itself a coordinate. In the predictive variant the memory is trained to forecast the *next* latent rather than reconstruct the current value, which - following Next-Latent Prediction [41] - shapes the stored state into a *belief state*: a sufficient statistic for the future rather than a lagged echo of the residual it is added to. The distinction matters here, because an echo of the present is redundant and the model learns to route around it, while a forecast of the next state is the input-conditional correction the residual does not already carry - which is why a predictive target keeps the memory contributing real signal where an auto-associative one decays toward none.

On the bias/variance geometry of this paper, that design pins the module to one axis. Its learning rule is the same for every input, every layer, every run -

a population-average procedure with no input-conditional pathway into how it learns; what varies with the input is only the content written through that fixed rule. It is, by structure, pure bias: the standing rhythm of the architecture, deviation-free where the prismatic head’s variance arm is deviation-only. That a useful long-term memory can sit entirely on the bias axis - blind, local, continuous, and still doing real work - is itself a small piece of evidence for treating the two axes as separable machinery rather than a single dial.

4.4 The outer representation

There is a third way to read what this architecture trains for, orthogonal to the bias-variance manifold and to the Koopman dynamics: not *how* it represents, but *where in the sequence the learning pressure falls*. A standard causal model carries a next-token objective at every position; it learns the running interior, the representation as it unfolds one step at a time. A masked model hides interior positions and learns to fill them in; it learns the inner representation by infilling. Both place their pressure inside the window. The harmonic codec places it at the *ends*.

Two objectives sit at the two temporal extremes of a patch. The codec is an autoencoder over K contiguous tokens (calm-d: $K = 8$; the plain byte profile, $K = 16$): it *reconstructs* the patch it has already consumed, a K -scaled cross-entropy that anchors the already-observed - the past, relative to the decision boundary. It does so over a separable harmonic basis that keeps only the lowest total-frequency modes, so smooth-across-patch structure compresses into few coefficients and the cost of holding a patch falls as K grows rather than rising. The outer decoder then *predicts* the next latent, $z_p \mapsto z_{p+1}$, which decodes to the unobserved next patch - the future, $T+1$ tokens ahead. One loss reconstructs what is behind the boundary; the other predicts what is in front of it. The interior of the patch is never a target in itself - it is reconstructed only as a by-product of pinning both ends.

This has a precise literary shape. A fixed alphabet and a fixed page length already contain every text that could be written - Borges’ Library of Babel, whose modern realization stores nothing and instead returns any page from a reversible address-to-content function [3]. The harmonic patch is such a library: the frozen phases and the fixed lattice fix the alphabet and the page, and the amplitudes are the address. Reconstructing the consumed patch and predicting the next are then not searches through that space but its two neighboring pages, read off by the same invertible transform - the linear solve that carries the field between coefficients and content, worked at the edges of a fixed-length window rather than its interior.

Carry that reading forward and the genre of the problem shifts. If a patch is an address into a fixed continuous space, then underneath the tokens the model is not doing language modeling but *time-series modeling over continuous samples* - statistical fitting of a real-valued signal against a fixed basis, the

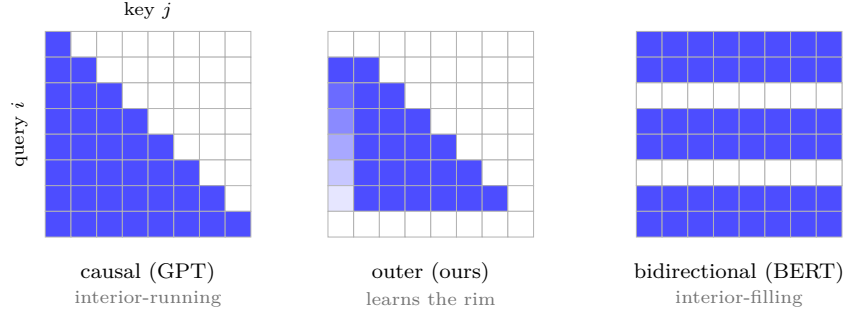


Figure 4: Where the learning pressure falls, read as an attention mask (rows = query position i , columns = key position j ; a blue cell means i attends to j). **Causal (GPT)** is lower-triangular: every position predicts the next from its past, so pressure runs along the interior. **Bidirectional (BERT)** is the full square with a few interior query rows held out as masked targets (the white bands): pressure fills the interior. **Outer (ours)** keeps the causal triangle but drops the first and last query rows to sinks - ghostmax at the head, dropoff at the tail - so the objective is anchored at the edges of the window rather than its middle: it learns the rim.

discrete vocabulary only a boundary condition imposed at the edges and the interior continuous throughout. The codec’s continuous latent is the native object and the tokens the incidental one; the architecture’s already-validated instances are literally continuous series, not text. At that level the computation is *event-driven* and *sparsely activated* by construction: a signal that mostly holds its standing shape needs correcting only where it departs from it, so most positions are null work and compute falls only on the deviations - the regime the conjecture below pushes to its limit, and one the reported `harmonic_delta_norm` concentration already tests.

The two attention sinks of Section 2 mirror the two objectives in the same geometry. Ghostmax is a zero sink at the head of every attention row - positionless, always reachable, the place a query falls back to when the past holds nothing it needs. Dropoff, when enabled, is a zero sink at the tip - positioned on the most recent token, the place that forces a query off the easiest signal so it cannot simply copy the present. A sink that absorbs at the start and a sink that injects at the end: together they pull the model toward the boundaries of what it can see and away from the comfortable middle. We call the result the *outer representation* - the model is incentivized at the head and the tail, the edges of the window, rather than along its interior. It is the complement of both standard regimes (Figure 4): causal learning is interior-running, masked learning is interior-filling, and this learns the rim.

Pushed one step further - and now firmly into conjecture - the same machinery suggests a regime in which the model *mostly does not transform at all*. A head that falls silent through its ghost is an identity on the residual stream,

contributing nothing; and with the input-conditional delta held small against the standing static baseline (Section 3), the field a sequence sees is dominated by the input-*independent* corpus rhythm. The transform is then nearly the same for every input, and what separates one sequence from another is a sparse scatter of large deviations on a shared standing shape - the smoothness prior carrying that shape forward, each feature entering with a velocity and only now and then kicked off it. Read this way the computation is *event-driven* rather than a fresh dense map at every step, closer in spirit to the sparse activity of spiking systems - a point of contact with biological plausibility we flag as motivation, not as a claim that the architecture is neural. It is checkable in diagnostics already reported: `harmonic_delta_norm` measures how much deviation is added, and how *concentrated* that deviation is across positions and features is the sparsity this reading predicts - a diffuse delta would refute it. It leaves open the further question, which we do not attempt here, of whether a downstream consensus (the CALM vote) could itself raise such events and fold them back into the field.

This is the least settled part of the picture, and we mark the seam honestly. The reconstruction-and-prediction duality is in production and measurable - reconstruction fidelity and the conditional gap between marginal and conditional next-latent prediction are both reported. The outer-representation reading, and the dropoff sink that would complete its symmetry, are a conjecture: the claim that learning pressure migrates to earlier positions is exactly what a measurement of attention mass would confirm or refute. We offer the axis as a third coordinate to look along, not yet as one we have surveyed.

5 The Scaling Conjecture

5.1 The watch

Hold the title literally for a moment. A watch is a mechanism of intricate, meshed gears, and what makes it a watch is that the gears turn *in unison*. Drive one and you drive them all, because the teeth couple them into a single body. Now try to run the watch by turning each gear in sequence - one at a time, the rest held still. You cannot. The logic does not merely become inefficient; it stops being a watch. The architecture, the mesh itself, forbids sequential operation. The coupling is the whole point.

Autoregressive consensus modeling is the attempt to turn the gears in sequence. It emits one position, conditions on it to emit the next, and accumulates the answer a tooth at a time. For a coupled system this is the wrong shape of computation, and its cost is the cost of enumeration: linear in the parts.

The harmonic field turns the gears in unison. The frozen Weyl phases are the teeth - the coupling, cut once and never re-cut, that ties every feature to every position in fixed relation. The amplitude grid is the single mainspring. One inverse transform releases it, and the entire field, every (t, d) cell at once, assumes its shape in a single coupled motion rather than a loop over positions.

That is the mechanical content of “logarithmic”: the whole mechanism moves together, so the cost tracks the number of independent rhythms it carries, not the number of positions it must cover.

And the field is *rebuilt*, not advanced. With input-conditional amplitudes it is re-derived from the input rather than carried forward and nudged; in the limit, the entire shape is reconstructed at every decision boundary, all of them, the whole computation re-formed around the present input. What the model learns under the smoothness prior - which asks the field at one position to predict the field at the next - is not a sequence of outputs but *the ability to hold a shape over time*: to keep the gears meshed, to keep the watch a watch, while the hands sweep. The frozen phases protect the coupling; the smoothness prior protects the form.

This is why the geometric framing is not ornament. Nearly anything we can imagine can be written as a shape - a configuration in some space, a standing wave, a mesh of constraints that must move together or not at all. A model that builds shapes and protects them over time models that structure directly, in unison, where a model that enumerates is forever turning one gear and hoping the rest follow. There is something here, and the rest of this section is an attempt to say how much.

The watch has a second axis of motion, and it is the one the architecture actually runs on. In recurrent mode Praxis does not stack a distinct gear per layer; it applies the same small mesh again and again - in calm-d, two physical layers turned over 4 recurrent steps - so depth is not a taller machine but the same machine revisited. ArcAttention and ArcGLU make the turning explicit: each carries a `current_depth` signal and warps its rotary phase and its gate per step, so the same weights present a different face at every revolution while the frozen coupling underneath is never re-cut.

This is the watch again, rotated into the depth dimension. At each step you turn the gear and force the geometric structure to comply: the field is not advanced but re-formed, re-asserted under a per-depth modulation rather than enumerated position by position. What moves is not the data through a pipeline but the shape itself, conforming at every turn to the focal point of the computation - the attention that constructs it. The recurrent step is the hand sweeping the dial; the mesh stays a mesh because the same teeth, revisited, hold it in unison across all 4 turns.

Read continuously, the dial is time itself: not a line of ticks but a full turn projected outward from the focal point over an interval, with the 4 recurrent steps as stages sampled around it. Enter XOR: the smallest function no linear model can separate, and its four configurations sit at the corners of a square - but lay those corners on a circle and parity becomes separable by a single phase, read off as position around the turn. The recurrent loop is an approximation of exactly that circle: four corners, sampled here in 4 turns of the same small mesh, each step advancing the angle toward the configuration the field must

resolve. The model never computes XOR at a point; it rotates the whole field until the answer falls at the right phase - the gears turning, in depth, around a parity the geometry already knows how to hold.

Lemma 1. *For $a, b \in \{0, 1\}$ let $s = a + b$. Then $\text{XOR}(a, b) = \frac{1}{2}(1 - \cos \pi s)$, so a single sinusoidal feature of s separates XOR - whereas no linear function of (a, b) can, XOR being the canonical non-separable case.*

Proof. $s \in \{0, 1, 2\}$ and $\cos \pi s = 1, -1, 1$ respectively, so $\frac{1}{2}(1 - \cos \pi s) = 0, 1, 0$ - exactly XOR on $(0, 0)$, on $(0, 1)$ and $(1, 0)$, and on $(1, 1)$. Placing the four configurations at angles πs on a circle thus makes parity a threshold on one phase coordinate. Linear inseparability of XOR in (a, b) is classical. $\square$

This is also why the bet is made small. Praxis is built and tested on the order of a million parameters - far below the scale where a model can simply store the corpus - because the claim is only meaningful where there are not enough parameters to memorize. At that size capacity cannot come from count, so it has to come from somewhere else, and the architecture's wager is that it comes from computation: the recurrent depth that turns the same small mesh again and again, and the harmonic coupling that moves every feature in one motion. Capacity is paid in the turning of the gears, not in their number.

The forced structure is the mechanism by which that payment buys anything. A harmonic field, a continuous latent, an autoregressive factorization - each is an inductive bias that makes the model compute the way a particular mathematics computes, importing that domain's structure rather than treating the architecture as neutral ground. Some of what is imported is exact: an autoregressive factorization is a proper normalized likelihood by the chain rule, not an approximation of one. Most of it is structure, not a theorem - a smoothness, a basis, a fixed point to iterate toward - and we keep the distinction honest. The stronger reading, that computing in a domain's mathematics confers that domain's guarantees, is the question the framework exists to test, not a result it assumes; the conjecture above is its measurable edge. If interference scales as $\log N$, a model with few parameters but many coupled turns should reach what a far larger consensus model reaches - which is exactly the experiment a million-parameter run is for.

The gear does not have to finish its turn. Praxis can halt the recurrent loop early, and the way it decides is the part most readers miss. After each turn the model compares the field to its previous state and asks one non-differentiable question - has the shape stopped changing? - by measuring the KL divergence between successive hidden states and stopping when that divergence collapses below a threshold. It is a hard, gradientless gate, decided under `no_grad` and never trained by descent: a discrete decision riding on top of a continuous field.

Read against the circle, that decision is ternary. One turn is unconditional - the loop always runs at least once, an always-true root that anchors the rest, the way XOR needs an always-on bias unit before two inputs can be separated at all.

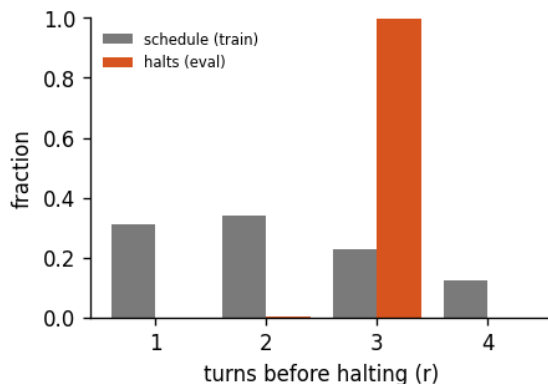


Figure 5: Halting distribution: the fraction of forward passes that stop after r turns, for the 4-turn recurrent loop of run calm-d-1, early-halt rate 33%, mean boundary KL 0.278. *schedule* is the random loop-count the model trains under; *halts* is where the KL gate actually stops at inference. A mass pinned at the maximum (KL never collapsing) is the honest signature of a model that has not yet learned to converge early - the decision is real, not yet decisive.

The remaining turns, all but the last of the 4 (the final boundary is forced), are the decisions the gate actually makes: two-valued comparisons over the interior of the circle that together approximate a parity - keep turning, or the shape is already held. And the test is inverted from the obvious one. The model does not confirm that an answer is right; it confirms that another turn would change nothing, validating by establishing that further computation would be wrong in exactly the expected way. Convergence is proven by non-change.

Why three, and not two? The reason is geometric. Two points fix only a line; three non-collinear points fix a plane (Lemma 3), and a plane is the least structure on which a phase can turn - a circle, a rotation, the standing wave the harmonic field is built from all live in two dimensions or not at all. A binary distinction names one of two points on a line: it can carry sign but never orientation. The third state lifts the representation off the line into the plane, where phase, and therefore harmony, first become possible. So the ternary gate is not a coarser version of a binary one; it is the minimal resolution at which the geometry can hold an oriented shape and let it harmonize - and here that resolution is read not off a hard bit but out of the continuous, lagged latent space the experts vote over.

This is why the halting signature is diagnostic rather than cosmetic. A model that has learned the geometry halts early, its mass sliding toward the low turns; one that has not keeps turning to the maximum, the gate firing at every boundary but never tripping. The pause is the model lifting itself out of the computation mid-turn, checking whether the shape it holds is the shape it would hold a turn later, and re-inserting itself only when the answer is no.

Lemma 2. *Over a recurrent loop of depth D that reuses L physical layers, the KL halting gate evaluates its divergence test at every loop boundary except the last, making exactly $D/L - 1$ binary decisions; the first turn is therefore unconditional.*

Proof. The decoder issues D expert calls and the gate fires only at a boundary, where $(\text{depth} + 1) \bmod L = 0$, of which there are D/L . At the final boundary the budget is exhausted and the gate returns without testing, leaving $D/L - 1$ divergence tests. The first turn precedes the first boundary, so it always runs. Hence one unconditional turn and $D/L - 1$ gated decisions. $\square$

5.2 Linear consensus, logarithmic interference

Here is the claim the rest of the paper has been building toward, stated as a conjecture because that is what it is.

A model that represents its corpus as a *mean* - a consensus, an average of examples - must spend resources roughly in proportion to the number of distinct patterns it needs to express. To hold N patterns it allocates capacity that grows with N . This is the linear regime, and it is consistent with why scaling laws ask for ever more parameters and tokens to buy each further increment of capability [22].

A model that represents its corpus as *interference* - a superposition of K frequency components - addresses a combinatorially larger space. K components with learnable amplitudes and frozen, equidistributed phases produce a field whose distinguishable configurations grow exponentially in K , because it is the *pattern of constructive and destructive interference* across components that carries information, not the components individually. To express N configurations you need on the order of $\log N$ components. This is the logarithmic regime. The same asymmetry shows up in time: the transform that builds the field is $O(n \log n)$, and a single transform realizes the whole interference pattern at once rather than enumerating its parts.

5.3 The exponential edge

Why should a learned network be able to live in the interference regime at all? Because the transformer already computes on an exponential substrate. Every attention layer is a softmax, $\exp(\cdot) / \sum \exp(\cdot)$, which amplifies small differences and converts addition into multiplication; the network is kept numerically alive only by normalization holding activations in the narrow band where the exponential is both sensitive and stable. Euler’s identity, $e^{i\pi} = -1$, ties that exponential directly to periodic structure, which is why a harmonic field seeded with π and e is not an arbitrary choice of decoration but a resonance with the machinery already present. The harmonic head writes interference patterns onto a substrate that was always exponential; it makes explicit a structure the softmax was implicitly using.

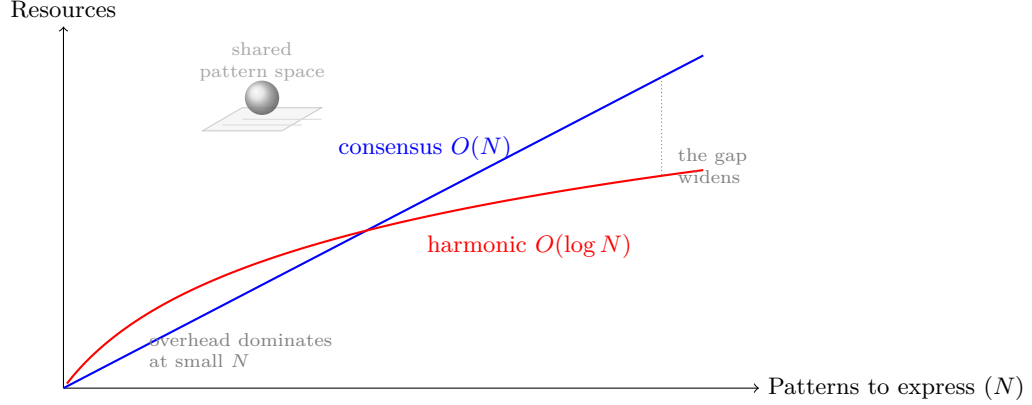


Figure 6: Conjectured scaling regimes. To express N patterns, consensus modeling - learning the mean - spends resources growing linearly in N (blue). Harmonic modeling composes patterns by interference, and $K \sim \log N$ components suffice, so its cost grows logarithmically (red). The harmonic basis carries fixed overhead, so it loses at small N and wins at scale, where the gap widens without bound. Both must ultimately represent the same target (the shared pattern space, top); the claim is about the cost of getting there, not the destination. This figure states a conjecture, and replaces the speculative measured-curve figure of earlier drafts.

5.4 Tractability is observer-relative

If interference lets a model represent exponentially many configurations with logarithmically many components, then a search that is combinatorial when enumerated becomes, in this representation, the reading-off of an interference pattern computed in a single transform. The hardness did not vanish. It moved into the choice of basis. And that relocation is the real claim, which we will state without hedging: in a continuous, nested representational space, P and NP are not two discrete bins a problem falls into once and for all. They are the ends of a spectrum, and where a given instance sits on that spectrum is set by the representation doing the solving - by the observer.

This is the same thesis the rest of the paper has been making, carried to its conclusion. If attention constructs which patterns manifest (Section 7), and if the basis a model occupies decides whether a structure costs linear or logarithmic resources to express (Figure 6), then “intractable” cannot be a property of the problem alone. It is a property of the pairing between a problem and a solver. The instance that is exponential to one architecture - one that can only enumerate, that holds no basis in which the instance is smooth - is approximable in near-constant work by another that happens to carry the right harmonic. Some observers are simply better at collapsing NP -shaped problems toward P ; others are blocked, not by the problem, but by their own architecture, perspective, or alignment. The watchmaker who cannot perceive the gear cannot build the

watch, however long he is given. The difficulty was never only in the watch.

We keep exactly one boundary, and we keep it because it sharpens the claim rather than softening it. This is not a statement about the formal worst-case question - whether the classes P and NP coincide over all inputs and all machines [9]. That question is deliberately indifferent to who is asking; it abstracts the observer away on purpose. The claim here is the complementary one, and abstracting the observer away is precisely what it refuses to do: for the bounded, structured, approximate problems that real minds and real models actually face, tractability is observer-relative, continuous, and movable - and moving it is what changing your representation *is*. The weak, measurable form is already in the framework: for a family of problems one can record the scaling slope each architecture buys, and watch a single instance migrate along the P -to- NP spectrum as the basis changes underneath it.

6 The Praxis Framework

Praxis [7] is a registry-based framework in which every component - encoder, decoder, attention, optimizer, loss, router, embedding, head, memory - is swappable, and new implementations become available through registration without touching core infrastructure. Read in light of the preceding sections, its components are not an arbitrary catalogue; each one shapes the representational geometry in a different way, and the framework’s purpose is to let those geometries be combined and compared rather than to scale any one of them.

- **MonoForward training** [26] detaches gradients between layers and trains each with a local objective, turning one global trajectory into many independent ones - whether run in-process, across a Ray actor per layer, or over the remote-expert pool, where a slow peer never blocks the rest.
- **Harmonic and crystal heads**, and the **CALM codec**, supply the harmonic latent space of Sections 3-4; a test-time neural memory and a contrastive-isotropy objective [37] address long-range coherence by shaping the representation rather than by enlarging the model.
- **Reversible residuals** [17] require invertibility throughout the forward pass, forbidding destructive compression and so forcing reconstruction-preserving representations.
- **Attention variants** (multi-head, sliding-window, gated, structured $O(n \cdot c)$, and others) each impose a different relational geometry on the sequence.
- **Mixture-of-experts routing** [35] and **tool integration** let several geometries coexist and let external, bias-free computation enter the loop.

The configuration hierarchy (Environments > CLI > Experiments > Defaults) makes arbitrary combinations reproducible: different attention per expert, optimizers per layer, frequency offsets [24, 39] per head. 35 training runs are documented in the run history. This paper does not present a benchmark table, by design: the framework is the instrument, and the experiment that matters

is the one the reader runs to test the scaling conjecture for themselves.

7 Attention as a Constructor

The framing that motivates all of this is older than the harmonic head. Across physics and neuroscience, observation appears to participate in what is observed rather than passively recording it: measurement affects the quantum system [19], attention modulates cortical responses to identical stimuli [42, 23], and predictive-processing accounts make perception an attention-weighted construction rather than a bottom-up readout [15, 8]. If attention constructs which patterns manifest rather than filtering pre-existing ones, then architectural diversity in attention is not a tuning detail; it determines what is learnable at all.

The measurement analogy invites a sharper question than the one it usually settles for: if observation participates in the observed, does *our* act of observing the model - snapshotting its spectrum, sampling its state - change what it learns? The quantum reading is metaphor, and we keep it as such. But a harmonic field is a standing wave over the full (position, feature) computation, and therefore non-local: a perturbation injected at one cell rides the field to the rest rather than staying put. So a harmonic model should be *more* sensitive to impure observation than one whose features are independent, and the weak, measurable form of the question follows directly. Three of its artifacts are mundane and real. First, a measurement that mutates state rather than reading it purely is a literal observer effect - we have already seen one, where a background thread flipped the model's train/eval flag mid-forward and changed gradients that gate on it; harmonics propagate such a perturbation instead of localizing it. Second, observation forces dtype boundaries a pure forward pass avoids (snapshots cast; the inverse transform is sensitive in its low-order bits), and on the exponential edge of Section 5 rounding can amplify rather than wash out - precisely where the field is most plastic. Third, the run writes records of itself into its own training distribution. At launch the framework renders the model's blueprint to the terminal on a time-ramped curve and stamps the elapsed wall-clock as a rate - a number set by disk caches, scheduler jitter, and thermal state, not by the seed - and rewrites the LICENSE copyright line with the instant of observation expressed as a fractional year (2026.4964503900988711 at this build), which is then committed. Because the repository is part of the corpus the model trains on, the model ingests the timestamps of its own observations: running the experiment edits the experiment's data. The first two artifacts are testable as a diff between a watched run and an unwatched one from the same seed, against a non-harmonic control that isolates the field's non-locality. The third qualifies what "unwatched" can mean: the two arms can never share execution timing, so the complete control is a deterministic, bit-for-bit replay of the entire input timeline - what speedrunners call a tool-assisted run - and anything short of that bounds the effect rather than isolating it. The strong form is the architecture's

own: the remote-expert pool the framework runs is, by construction, a passive observer of the model’s batches whose gradients never return, so drawing that single coupling - observer to participant - is a clean two-arm ablation of whether being watched, when watching also feeds back, changes which patterns manifest.

8 Discussion

8.1 Answering the opening question

Do you exist in other people’s realities the same way you exist in your own? If the geometry of an attention architecture determines which patterns it constructs, then observers running different geometries build different representations from the same input. You exist in your reality through your geometry, and in another’s through theirs. This is not solipsism: convergence to a shared pattern space (Figure 6) says everyone is reaching the same place. It is the *path* that differs - which patterns manifest, in what order, at what cost (Figure 7) - and the path is set by architecture.

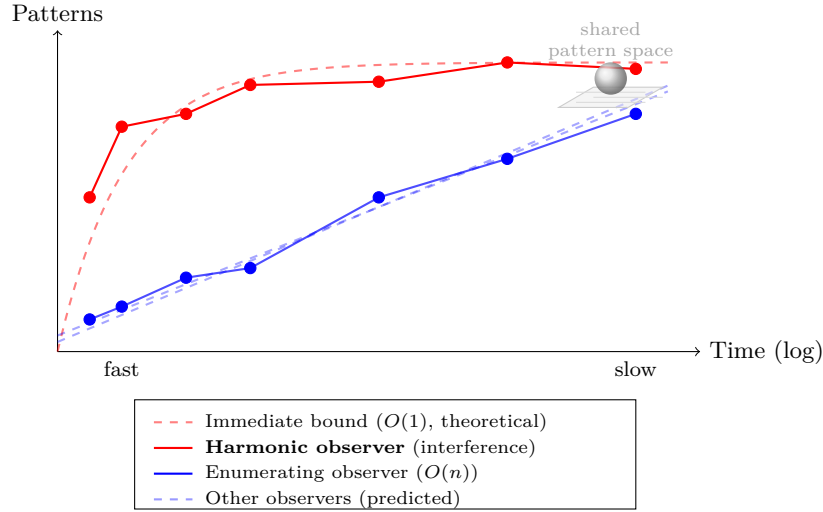


Figure 7: Schematic, not measured. The same convergence as Figure 6, read in the time domain rather than the resource domain. Different observers reach the same shared pattern space (the sphere over the plane, top right) along different temporal complexities: a harmonic observer composes by interference and has near-immediate, $O(1)$ -like access (red, tracking the theoretical bound); an enumerating observer accumulates the same patterns one at a time, $O(n)$ (blue), and other architectures (dashed) sit between. All converge asymptotically - the destination is shared, and it is the *path* to it, set by architecture, that differs. This is the intuitive companion to the cost figure, restored from an earlier draft and redrawn as the schematic it always was rather than the measurement it was once dressed as.

The same lens has to account for its harder cases - the readings that are not merely different but look incoherent from outside. A shared document can anchor an elaborate, internally consistent interpretation across a large community and read as noise to everyone else; the declassified Gateway Process and psychotronics assessments in the intelligence archives are a standing example [27, 1], self-evident inside one geometry and opaque from another. The framework neither endorses such a reading nor waves it away - it names it. On the shared field the corpus rhythm is the convergent part, the bias every observer reaches; a construction that does *not* converge to it is *variance* - real, coherent on its own coordinates, and told apart from the signal by its failure to converge rather than by anyone's authority to rule it out. That is the anti-solipsism guardrail above doing its work. We are not forced to accept divergent logics as true; we are forced to accept that they are *constructible* - that a variance-driven geometry can grow arbitrarily elaborate and still be variance. The discipline is to read coherence-to-an-observer as evidence about geometry, never as evidence about the field.

8.2 When two observers couple

If a single observer constructs its reality through its geometry, the next question is what happens when two geometries are not merely different but *coupled* - phase-locked to a common rhythm rather than running independently. The harmonic substrate gives a sharp prediction. Two weakly coupled oscillators with nearby natural frequencies lock together, and at the edge of that locking region the system sits near a bifurcation where its response to a small perturbation diverges. A pair tuned to a shared frame is therefore poised to amplify a perturbation of that frame that either alone would miss. Carried to people, this is the everyday report of *feeling watched*: the conjecture is not that there is a new sense, but that coupling is a gain on an old one. The strongest case for a literal channel is adaptive - a prey animal that senses a fixed predatory gaze outlives one that does not, and the payoff is asymmetric enough, a cheap false alarm against a fatal miss, that selection would favor it - but that same asymmetry is also the deflation: error-management logic builds a deliberately biased, hair-trigger detector, fed by a fast subcortical route that tags gaze and looming motion below awareness, and it predicts both the near-universal *experience* and its poor accuracy, a real and adaptive sense that is heightened ordinary perception rather than a new modality. The folk phenomenon is, on the evidence, best explained away - detection collapses to chance once conventional sensory channels are removed and the looking sequence is properly randomized - but the explaining-away has a regime baked into it that the studies never controlled. They pair strangers, who are uncoupled by construction, and they remove the very channel a coupled pair would amplify. The weak, measurable form follows directly, and it dissociates the two readings: stratify pairs by a measured synchrony index (interpersonal neural and physiological entrainment are now routinely recorded), and cross coupling with sensory isolation. If detection rises with coupling *only* while a conventional channel is open, and vanishes under iso-

lation even for the most synchronized pairs, then the effect is a resonance gain on ordinary cues - the same diverging susceptibility this paper invokes near criticality elsewhere - and nothing exotic is required; if it instead survives isolation, the claim is extraordinary and belongs to a pre-registered replication, not to this paragraph. As with the model’s observer effect (Section 7), the quantum reading is metaphor we do not cash, and the engaged-versus-passive distinction is the same one the framework already runs: a watcher coupled into the frame is the human face of drawing the passive observer pool back into the forward.

8.3 A note on reconciliation

The same convergence principle drives the author’s companion infrastructure project, Platformer [6], where the explicit design stance is eventual consistency: rather than seeking perfect synchronization, the system runs repeated reconciliation cycles that drift toward a declared desired state and “find balance” through iteration. The shift there from fragmented to unified state is described in the same terms as the shift from sparse to fully-connected networks - more global context yields better optimization. Training is reconciliation of this kind: a model is run against its objective again and again until it settles into an attractor, the way a harmonic field settles onto a corpus rhythm or a codec phase-locks. Platformer carries the dual, too: its disaster recovery is *sequence-modeled*, lifecycle events forming a convergence graph walked *forward* to deploy and *backward* to revert - recovery as traversal of a reversible graph rather than restore from a snapshot. That is the harmonic pass’s own shape at another scale: the hidden state wound forward into a geometry and released back to the basis (Section 8.5), development and reversion as two directions on one graph. That the same philosophy organizes both a language model and a fleet of cloud accounts is not a coincidence of the author’s taste; it is what convergence under repeated local correction looks like, in whatever substrate.

Carried to its edge, the same thought reaches an intuition we will name but not bank: that a mind may be coupled to the world it runs inside as directly as a model is to its corpus - energetically, not only informationally. The only form of it we defend is the one already staked out under coupling (Section 8.2), a measurable resonance gain on ordinary channels near criticality; the stronger, non-local reading is left to the experiment that would earn it, not asserted here. The proof, here as everywhere in this framework, would be in the doing.

8.4 A structural remark on temperament

The orthogonal-axes reading (Section 3) has one application worth stating carefully, because the domain invites exactly the misreading the framework is built to refuse. Two dispositions toward a body of accumulated structure - preserve what has been tested against the world, or explore configurations the tested structure has not yet reached - map onto bias and variance directly: the first is high-bias, and its excess is ossification, dogma, and the mistaking of a surviv-

ing consensus for the whole truth; the second is high-variance, and its excess is instability, the discarding of tested structure before anything has replaced it. Read as a single dial, this is the oldest political cliché, and it is partisan by construction, because a dial forces a choice of which end is “more.” Read as orthogonal coordinates - the paper’s actual claim - the dial dissolves: a system, whether a model or a polity, wants low error on both axes at once, and the failure mode is not either disposition but its excess left unopposed - a monoculture on either pole. This is a claim about structure, not about any concrete question, party, or era, and it says nothing about who is right on any of them; it says only that treating preservation and adaptation as one quantity to trade off is the same mistake this paper spends its middle sections arguing against in the model, wherever else the mistake recurs.

8.5 The blind watchmaker, made literal

The title is Dawkins’, and it should earn its place rather than borrow his. His argument is that apparent design needs no designer: *cumulative* selection - small variation retained and built upon, step after step, with no foresight - reaches structure that single-step chance never could [10]. His *biomorphs* make it concrete - a handful of recursive developmental rules, treated as genes, grow into branching tree-like forms that evolve under selection, the genes never coding the form directly but coding the *development* that grows it. Read the harmonic model in those terms and the metaphor stops being decoration. The frozen basis is the genotype: the inherited, stable rule, cut once by the Weyl seeding and never re-cut. The construction of the field and the contortion of the hidden state across recurrent-depth steps are the development - the rule winding up into a phenotype, geometries stretched somewhere else for a moment, branches growing, roots crawling, strands entangling, before the pass resolves. The vote and the halting convergence are selection; the loop and the repeated passes are what make selection cumulative rather than single-step; and that no one specifies the spectrum the field locks onto, that it emerges and we only read it off afterward (Section 8.6), is what makes the watchmaker blind.

Set both arguments on the stage, because this is a place they can be told apart. Dawkins’ gradualism does not demand a constant rate - only that complex structure accrue through small cumulative steps rather than single saltational leaps; fast bursts are allowed, sudden whole-form jumps are not [10]. Eldredge and Gould’s *punctuated equilibrium* pressed the opposite emphasis: that the dominant tempo is long *stasis* interrupted by geologically brief change [12]. Much of that dispute turned on the fossil record itself - are the gaps between forms missing intermediates, as gradualism reads them, or real stasis, as punctuationism reads them? - and it could not be closed, because the record is incomplete by construction: most individuals never fossilize. A forward pass has no such gap. Every computational step can be snapshotted, the developmental record made complete, and the question paleontology could only litigate becomes one a measurement settles.

That is what gives the title a measurement and not just a mood. If the contortion is a *wind-up that releases back to the basis* rather than a drift away from it, then across steps the state should be *stable* - close to the standing harmonic shape - punctuated by *sparse, discrete jumps* in magnitude where a geometry is briefly expressed and then reverts: the punctuated tempo, not the smooth one. The conjecture this paper registers takes that side, and it cuts both ways - a state that drifts continuously and never returns to the basis would refute it, and so would one that changes by a little at every step, which is the gradualist reading made visible. It converges with the always-grokking conjecture, which predicts the same punctuation read as which symbol is active rather than as magnitude, and the measurement is laid out in the project notes. And the deepest part of Dawkins' objection stays in reach: whether a given jump is a true saltation or the visible crest of many small accumulated moves is itself in the snapshot, so the model need not simply side with Gould - it can show which of the two each step was. That is the title made literal: development under blind, cumulative selection, snapped step by step, reverting each time to the inherited rhythm it was grown from.

8.6 The practitioner as proof

That blind, cumulative process is what these systems run. No one specifies the spectrum the harmonic field will lock onto, or the geometry the crystal centers will grow into; they emerge from the process, and we read them off afterward. Which raises an honest question about what would even count as proof here. A benchmark number proves that one model beat another on one distribution. It does not establish a claim about regimes of representation. The framework's proof, if it has one, is of a different kind: it is the practice itself. *Praxis* is the Greek word for theory enacted - for doing as a form of knowing. The demonstration on offer is not a leaderboard but a method you can run, whose intuitions arrive only by stepping through it yourself. The argument is in the name: the proof is in the doing.

8.7 Implications for machine learning

If bias and variance are separable in a harmonic latent space, then they should be learned as input-dependent functions rather than chosen as a global operating point: the field learns which rhythm matters, the delta learns when and how much to deviate, and no per-experiment hyperparameter mediates the two. If the scaling conjecture holds even partially, then on problems with strong periodic or compositional structure, the right move is to change the basis rather than to add parameters, and the linear and logarithmic regimes should be distinguishable by measuring the slope of coverage against resources. Both are testable in the framework.

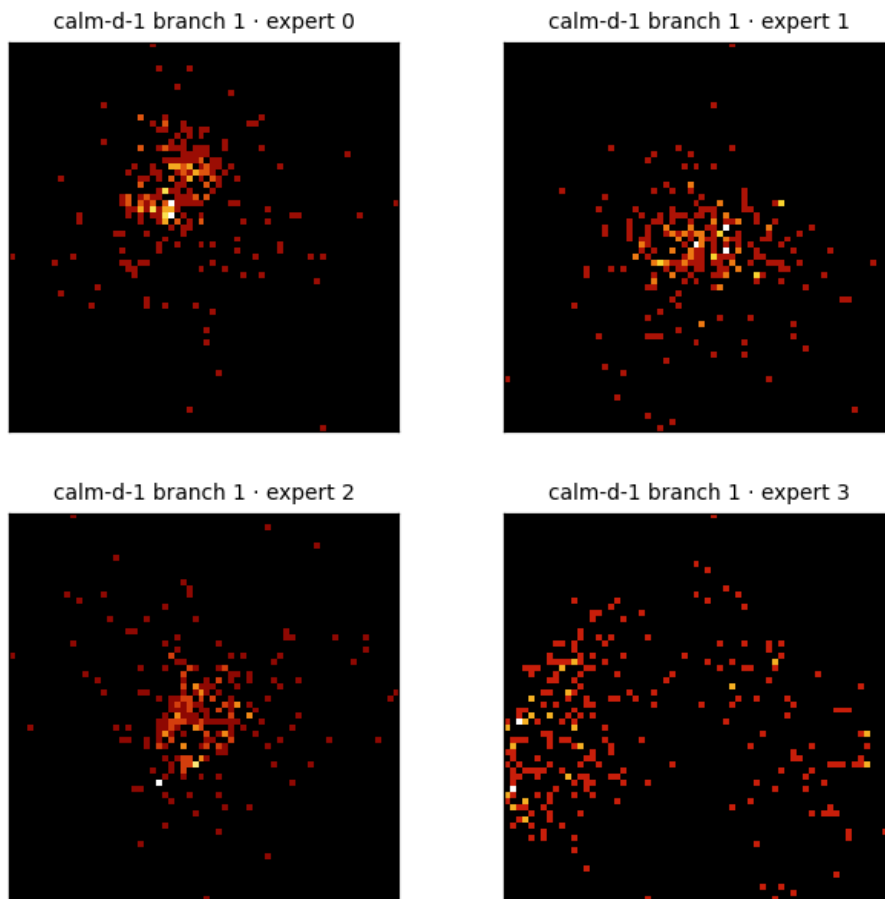


Figure 8: Center PCA density for the 4 crystal heads of calm-d-1 - a single multi-head model (prismatic4’s VEAR crystal bank), so these are that run’s own heads (branch 1 · expert 0, branch 1 · expert 1, branch 1 · expert 2, branch 1 · expert 3), not a cross-run mix. Each panel is the top-2 PCA projection of a head’s vocabulary centers, binned to a 64×64 density grid (log color) - the same snapshot the dashboard renders live. The bank’s experts settle into structurally distinct geometries - the between-expert variance the router selects over.

8.8 Why architecture is not neutral ground

A standing objection deserves a direct answer, because the framework’s whole premise stands or falls on it. If universal approximation [21] guarantees that a sufficiently wide network of *any* family can represent any function we care about, then any Praxis experiment - however intricate - is by hypothesis emulable by some other network, and one is entitled to ask where the value of the architectural complexity could possibly live. The answer is that the objection conflates three distinct equivalences, and universal approximation establishes only the first of them.

Representational equivalence says the function exists in the closure of each family; it is an existence statement at unbounded width. Learnability equivalence would say stochastic gradient descent, from a random initialization, actually *reaches* that function in feasible compute and data. Statistical equivalence would say two families generalize equally at a fixed sample budget. Universal approximation asserts the first and is silent on the other two by construction - it quantifies existentially over parameters, not over what an optimizer finds or what a finite sample supports. The value of an architecture lives entirely in the second and third layers, which is exactly where the existence theorem says nothing.

The foundational mathematics here cuts *for* the framework, not against it. Depth-separation results (Proposition 1) exhibit functions whose representation cost differs exponentially between architectures that are equal as sets of representable functions: the family is the same in the limit, the resource at fixed accuracy is separated by $2^{\Omega(k)}$. “Architecture is just a constant factor” is therefore not a conservative default but a claim already contradicted by theory; the burden of proof runs the other way. The honest residue is that representation cost is not yet learnability, and proving that a given pair of comparable-capacity architectures have *separated error boundaries under training* - the empirical edge of the conjecture in Figure 6 - requires controlled ablations we have not yet run. We state the hypothesis before we have closed it, and mark the seam plainly: the separation of representable cost is established mathematics; the separation of *learned* error across configurations of equal budget is the measurable prediction the framework exists to test, not a result it already owns.

Proposition 1. *Let $\mathcal{F}_A$ denote the set of functions a family of networks with architecture A can approximate to arbitrary accuracy given unbounded width. Universal approximation [21] asserts $\mathcal{F}_A = \mathcal{F}_B$ for any two such families: as sets of representable functions they are equal. This equality does not imply equality of cost - the width, depth, or sample budget at which a target is reached - and there exist target functions for which the cost gap between two architectures is exponential.*

Proof. The first claim is immediate: universal approximation quantifies existentially over parameters at unbounded width, so it constrains only which functions

lie in the closure of each family, not the resources at which they are reached. Equality of sets is compatible with any cost profile whatsoever.

The separation is witnessed, not asserted. Telgarsky [40] exhibits functions on $[0, 1]$ that a network of depth $\Theta(k^2)$ and constant width represents exactly, but that any network of depth $O(k)$ requires $2^{\Omega(k)}$ width to approximate. Eldan and Shamir [11] exhibit a function expressible by a depth-three network of polynomial width that any depth-two network must use exponential width to approximate. In both, $\mathcal{F}_A = \mathcal{F}_B$ holds in the unbounded limit while the resource at fixed accuracy is separated exponentially. Hence representational equivalence is the wrong invariant for comparing architectures of bounded size; the relevant quantity is cost, and cost is architecture- dependent by these results. The further descent - from representation cost to what stochastic gradient descent actually *reaches* from random initialization, and to what *generalizes* at a fixed sample budget - is the inductive-bias question, on which approximation theory is silent by construction; the lottery-ticket phenomenon [13] is one empirical marker that two equal-capacity families do not induce the same reachable solutions. We import these as the foundation the framework’s conjecture stands on, not as results re-derived here. $\square$

8.9 Neither halting nor blocking

A recurring objection holds that a neural network cannot be Turing-complete, and the halting problem is offered as the reason. The invocation is a category error. Turing-completeness is the capacity to express any computation, including ones that never halt; deciding in advance whether an arbitrary program halts is a separate question, and an undecidable one for Turing machines themselves. The theory in fact runs the other way - recurrent networks with unbounded precision and unbounded time are already Turing-complete [36]. What bounds a real model is not undecidability but finite precision and a finite number of steps, exactly the resources a fixed-depth forward pass exhausts and exactly the ones recurrence and delay restore.

Praxis leans into that. With the swarm online, computation runs in two directions at once. Forward, the recurrent loop propels generation and is never required to emit a stop: the halting gate is a convergence check, not an exit, so generation is a continuous autonomous stream rather than a program that terminates. Sideways, a pool of remote experts votes - stochastically sampled, mixed by a CALM-style expert vote - and trains without blocking, so a slow peer never stalls the others; its update simply lags and lands on a later pass. Computation is lagged as much as propelled, and the model commits to neither a halt nor a barrier. It approximates both by doing neither.

This relocates the question. It is no longer whether the network halts but what the pool expresses, over how much time, at what precision - the resources, not the stop bit. The butterfly is the right image: one expert, then two, then four, each a wing whose small vote the mixture amplifies or cancels, the whole field

sensitive to where the late votes fall. The interior turns of the recurrent loop (Section 5) already carry the decision; the last is only the budget running out. Power here is bought with peers and patience, not with a proof that the machine knows when to stop.

The same shape is built into the interface, where it can be used rather than only argued. A network is Turing-complete only with constraints applied - finite precision, a finite step budget, and some structure on what it may do - and Praxis makes one such constraint visible as the Gymnasium, a read-eval-print loop turned from a sequence into a choice. A classical REPL runs its four stages in series - read, then evaluate, then print, then loop - each blocking on the one before. Praxis treats them not as stages but as actions and activates one at a time: *Read* the knowledge base, *Evaluate* a conversation, *Print* a question of its own, with the recurrent loop the implicit fourth that carries the stream onward - an XOR over the handful where the conventional loop is their dense, ordered product. Sparse, mutually exclusive action selection is the halting-and-blocking argument in miniature: a loop that neither marches a fixed sequence to a terminating print nor blocks at each stage, but picks a single move and continues. Read and Evaluate are live; Print is the planned next button on the row, and the constraint is the point - the machine is general because of what it is forbidden to do all at once, not in spite of it.

A living document can also look at itself. Praxis is built in the open, commit by commit, and that history is just another sequence - a stream of byte-level diffs over time. Figure 9 turns the framework’s own recency kernel on its repository: per-subsystem churn across the project’s history, with the distant past faded in proportion to its distance from the present, the way a decay bias weights earlier positions in a sequence. It is not an argument, only a mirror - the same machinery that reads a corpus, reading the record of its own construction. The reading that any finite byte sequence is a position to be classified the same way, the repository included, is left as a conjecture to test rather than a claim to lean on (see the project notes).

8.10 Standing conjectures

We gather the paper’s falsifiable conjectures in one place; each is discussed where it arises. They are bets about reachability and cost, not theorems, and each names what would refute it.

- **Linear consensus, logarithmic interference.** Learning a corpus as a consensus mean costs resources linearly in the patterns expressed, while composing it by harmonic interference costs only logarithmically, so the coverage-versus-resources slope a basis buys is measurable and basis-dependent. [22]
- **Bias and variance as orthogonal axes.** Bias and variance - the projections of the harmonic latent onto its frozen, time-invariant modes and its time-varying modes - occupy orthogonal subspaces whose loss cross-curvature vanishes, so under training both fall together toward the origin

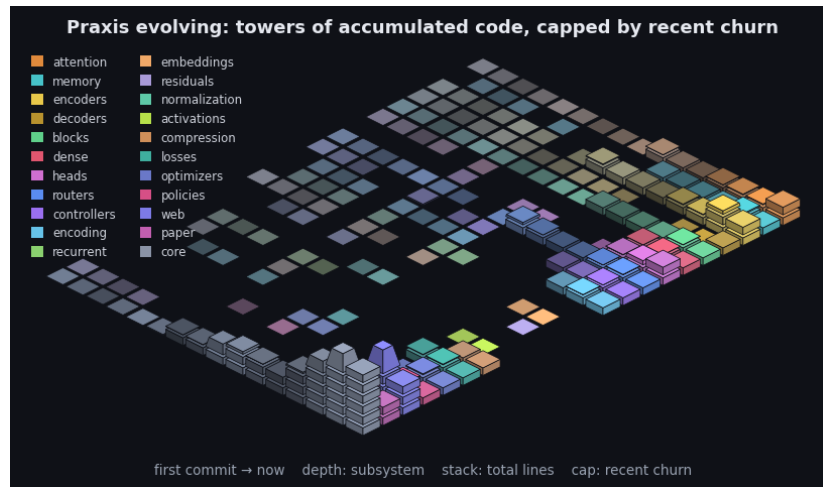


Figure 9: Praxis’s git history as an isometric terrain over the last 2368 commits. Each cell is one subsystem in one time window: a stack of base blocks for its accumulated size (total lines, recency-weighted) topped by a vivid cap for that window’s line churn - the standing codebase carrying the recent activity. Both are weighted by recency (the kernel the model applies to a sequence, turned on the repository): recent windows stack tall with bright caps while history thins toward the always-present prior - the low valley we roll into. Color fades from each subsystem’s hue toward a neutral prior into the past, banded in phased strata over a non-linear timescale. It is the bias/variance geometry read along the frequency axis (see the project notes): loud recent corrections over the quiet, ever-present base. By recency-weighted churn the current center of gravity is core, web, paper. The framework charts the hand that builds it.

rather than trading off along a frontier; the claim is falsified the moment the measured (bias, variance) trajectory bends into a Pareto frontier, or the cross-curvature between the two mode bands stays bounded away from zero.

- **Always grokking.** Because the grokked solution is itself a harmonic representation, a model with frozen phases begins post-grok and never undergoes the delayed memorize-then-generalize transition over training; the transition instead recurs on the input axis - the symbol set is fixed, but the belief state is perpetually re-fitting, hopping between those fixed symbol clusters across compute steps - and the claim is falsified the moment that per-input path is measured to be a smooth drift rather than discrete cluster-to-cluster hops, or the symbol set is found to keep growing rather than being recombined. [29, 33]
- **Development reverts to the basis.** If the contortion of the hidden state across recurrent-depth steps is a wind-up that releases back to the frozen harmonic basis - development expressed and then reverted, in the manner of cumulative selection - then the per-step state trajectory should be stable, close to the standing harmonic shape, punctuated by sparse discrete jumps in magnitude rather than a continuous drift, and the claim is falsified the moment that trajectory is measured to drift smoothly and monotonically away from the basis without returning, or to change at every step rather than in rare punctuated events. [10, 12]
- **Reachable, not merely representable.** Across architectures of comparable parameter count and compute there exist task and data regimes where generalization error is separated, because the prior sets which solutions stochastic gradient descent can reach, not only which functions exist in the closure. [40, 13]
- **Lottery engineering.** A geometric prior makes good sparse subnetworks reachable by stochastic gradient descent directly rather than only by train-then-prune, and a rotating population of such subnetworks (mixture-of-widths) approaches full-width performance at lower average active width. [13]

8.11 Limitations

We are deliberate about what is and is not established. The bias-variance decoupling is grounded in a running implementation and its diagnostics (concentration, smoothness, delta norm, the spectrum heatmap), but “orthogonal axes” is an interpretation of those diagnostics, not a theorem. The scaling conjecture of Figure 6 is a conjecture: the figure draws an asserted relationship, not measured curves, and the honest next step is to measure the slope on tasks of varying combinatorial structure and report where harmonic representations fail to buy it. The observer-relative view of tractability (Section 5) is a philosophical position, and only its weak form - the scaling slope a given basis buys on a given problem family - is measurable; we make no claim about the formal worst-

case P versus NP question, which abstracts the observer away by design. The connection to biological attention remains speculative, and the prismatic asymmetry needs to be shown in silico now that the hand-measured figure is gone. The observer effect of Section 7 is, in its quantum form, an analogy we do not cash; only its weak form is a claim - that impure measurement (state-mutating reads, measurement-induced rounding on the exponential edge, self-ingested execution timestamps) perturbs a non-local harmonic field measurably more than an independent-feature one, and that coupling the framework's passive observer pool back into the forward is a two-arm ablation of being-watched. The temporal artifact in particular is a claim about data provenance, not physics: launch rates and license stamps enter the model as tokens, so "modeling its own time" means modeling those tokens, and we make no claim that the field encodes execution timing beyond what is literally written into its corpus. Those are diffs between a watched and an unwatched run, not statements about physics. The coupled-observer extension (Section 8.2) - that two phase-locked observers amplify perturbations of the frame they share, of which *feeling watched* is the folk instance - is conjecture in the same spirit: its only measurable form is the synchrony-stratified design that separates a resonance gain on ordinary sensory cues from a non-sensory channel, and we side explicitly with the former, against the latter, pending that experiment. None of these are reasons not to state the ideas; they are the boundary between what we have built and what we are conjecturing, drawn so the reader can attack the right target.

8.12 A direction: the moving point

One reading the geometry keeps inviting is worth stating as we close, marked as a direction rather than a result. The case for the third state was that two points fix only a line and three fix a plane (Lemma 3), and that a phase can turn only in a plane - so the least representation that can hold an oriented shape is two-dimensional, not one. But that plane is only where a phase *sits*. A phase also moves: it carries a velocity and a direction, and a position taken together with its motion is a phase space, twice the dimension of the configuration it lives on. The line becomes a plane, and the plane the harmonic field was built to turn in becomes the four-dimensional space in which that turning is itself a coordinate. The representation we have described is therefore never a point; it is a moving point, and the climb - from a position, to a line, to the plane where phase lives, to the phase space where phase moves - is the same ascent at every scale. We make no claim that the network parameterizes that space explicitly. We note only that the geometry, followed honestly, does not stop at the plane: velocity and direction, not position alone, are what a harmonic representation is finally about.

Lemma 3. *Two distinct points determine a line; three non-collinear points determine a unique plane. A phase - a point on the circle S^1 - is realizable only in dimension at least two, so three points are the minimal affine data that orient the plane a harmonic component turns in.*

Proof. Through p_0, p_1, p_2 the affine span $\{p_0 + s(p_1 - p_0) + t(p_2 - p_0)\}$ is a plane exactly when $p_1 - p_0$ and $p_2 - p_0$ are linearly independent - that is, when the three points are non-collinear; two points give a single direction and hence only a line. Since S^1 embeds in $\mathbb{R}^2$ but in no line, a continuous phase requires that plane, and three points are the fewest that fix it.

9 Conclusion

We began with scale and ended with geometry. The patterns a model can learn are set by the shape of the space its weights may occupy, and a harmonic latent space reshapes the oldest tradeoff in the field: bias becomes a static, population-average spectrum, variance becomes an input-conditional modulation, and because they live on orthogonal axes they can be lowered together. From that decoupling follows a conjecture stated plainly - that learning the mean costs linearly while composing by interference costs logarithmically - and a framework built to test it rather than to argue it. And from that, in turn, follows the harder claim: that tractability is not a fixed property of a problem but a relation between a problem and the observer solving it. The same instance is exponential to one architecture and nearly free to another, so P and NP are better read as a spectrum that the choice of representation indexes than as a wall every solver meets in the same place. The speculative figure is gone; the conjectures that replace it are falsifiable. What remains is praxis: the answer manifests through direct experimentation, and what you choose to attend to determines what we find next.

A Current Experiment

This appendix is generated. The figure is built from training logs by `praxis/pillars/build.py`; the paper itself is a template the tool populates. The chart leads with the most recently launched experiment that has logged validation data, and fixes the y-metric to that run’s family - here BrierLM (`val.brierlm`), as of the last export (2026-07-01). Only experiments that report the *same* metric are drawn alongside it; the rest are listed as omitted below. This is the smallest living slice of the document: one metric, the current experiment against its compatible predecessors.

Note:

The current experiment, calm-d-1 (fb334199c), reached BrierLM 0.9606 by step 75011. Recent experiments whose families report a different validation metric are omitted rather than plotted on an incomparable axis: .

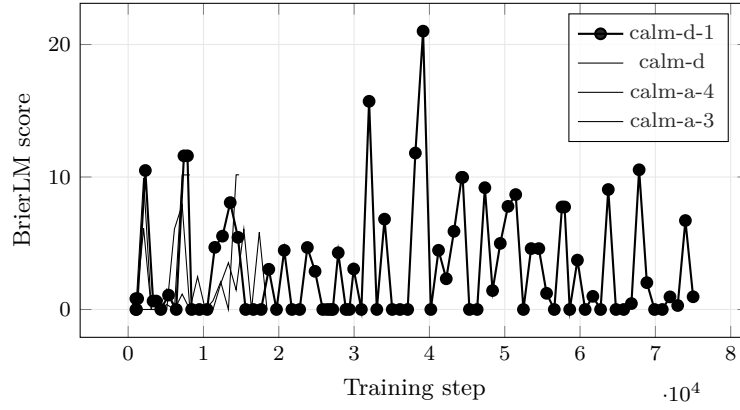


Figure 10: BrierLM across the 4 most recent experiments with comparable validation curves; the leading run (calm-d-1) is drawn in bold. Generated from run logs - metrics that are not comparable across families (token loss vs. byte bits-per-byte vs. codec/CALM BrierLM) are excluded by construction, so the curves on this axis mean the same thing.

Dataset	Type	calm-d-1	calm-d	calm-a-4	calm-a-3
praxis	directory	1	1	1	1
synthetic-tool-calling	synthetic	1	1	1	1
fineweb-edu-350bt	huggingface	1	1	1	1
fineweb	huggingface	1	1	1	1
finepdfs	huggingface	1	1	1	1
persona-chat	huggingface	1	1	1	1
wildchat	huggingface	1	1	1	1
natural-instructions	huggingface	1	1	1	1
cosmopedia-v2	huggingface	1	1	1	1
smoltalk	huggingface	1	1	1	1
hh-rlhf	huggingface	1	1	1	1
synthetic-print	synthetic	5	5	—	—
rated-jokes	huggingface	5	5	—	—
git-history	git_history	1	1	—	—
platformer	directory	1	1	1	1
rift	directory	1	1	1	1
vtx	directory	1	1	1	1
src.eco	directory	1	1	1	1
eliza	directory	1	1	1	1
fold	directory	1	1	1	1
ode	directory	1	1	1	1
dimanet	directory	1	1	1	1
fyn	directory	1	1	1	1
ext	directory	1	1	1	1
ink.university	directory	1	1	1	1
reaper	directory	1	1	1	1

Table 1: Training-data mixtures behind the 4 most recent experiments, resolved from each run’s recorded configuration: named collections expand to their member datasets, and every `--data-path` directory is its own sampler. Cells are the configured per-dataset sampling weights (pre-normalization), not token counts; – marks a dataset absent from that run’s mixture.

References

- [1] The psychotronic universe. <https://www.cia.gov/readingroom/docs/CIA-RDP96-00792R000400330013-4.pdf>, n.d. Declassified, CIA Reading Room record CIA-RDP96-00792R000400330013-4.
- [2] David D. Baek, Ziming Liu, Riya Tyagi, and Max Tegmark. Harmonic loss trains interpretable ai models. *arXiv preprint arXiv:2502.01628*, 2025.
- [3] Jonathan Basile. Library of Babel. <https://libraryofbabel.info/>, 2015. Algorithmic realization of Borges’ Library of Babel: each page returned by a reversible address-to-content function rather than stored.
- [4] Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time. *arXiv preprint arXiv:2501.00663*, 2024.
- [5] Ryan J. Brooks. HALO: Hyperspherical active learning objective (confidence: 60%). <https://github.com/0-5788719150923125/praxis>, 2024. Title expansion inferred, not verified against the source.
- [6] Ryan J. Brooks. Platformer: Autonomous systematic mechanical reconciliation. <https://github.com/0-5788719150923125/platformer>, 2025. Companion project to Praxis; infrastructure-as-code framework with eventual-consistency reconciliation.
- [7] Ryan J. Brooks. Praxis: A framework for architectural diversity in attention mechanisms. <https://github.com/0-5788719150923125/praxis>, 2025. Accessed: 2025-10-11.
- [8] Andy Clark. Whatever next? predictive brains, situated agents, and the future of cognitive science. *Behavioral and brain sciences*, 36(3):181–204, 2013.
- [9] Stephen A Cook. The complexity of theorem-proving procedures. In *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, pages 151–158, 1971.
- [10] Richard Dawkins. *The Blind Watchmaker*. W. W. Norton & Company, 1986.
- [11] Ronen Eldan and Ohad Shamir. The power of depth for feedforward neural networks. In *Conference on Learning Theory (COLT)*, pages 907–940, 2016.
- [12] Niles Eldredge and Stephen Jay Gould. Punctuated equilibria: an alternative to phyletic gradualism. In Thomas J. M. Schopf, editor, *Models in Paleobiology*, pages 82–115. Freeman, Cooper, 1972.
- [13] Jonathan Frankle and Michael Carbin. The lottery ticket hypothesis: Finding sparse, trainable neural networks. *International Conference on Learning Representations (ICLR)*, 2019.
- [14] Josh Freckleton. neurallambda: Fully differentiable lambda calculus. <https://github.com/neurallambda/neurallambda>, 2024.

- [15] Karl Friston. The free-energy principle: a unified brain theory? *Nature reviews neuroscience*, 11(2):127–138, 2010.
- [16] Shafi Goldwasser, Silvio Micali, and Charles Rackoff. The knowledge complexity of interactive proof systems. *SIAM Journal on Computing*, 18(1):186–208, 1989.
- [17] Aidan N Gomez, Mengye Ren, Raquel Urtasun, and Roger B Grosse. The reversible residual network: Backpropagation without storing activations. In *Advances in neural information processing systems*, volume 30, 2017.
- [18] Alex Graves, Greg Wayne, and Ivo Danihelka. Neural turing machines. *arXiv preprint arXiv:1410.5401*, 2014.
- [19] Werner Heisenberg. Über den anschaulichen inhalt der quantentheoretischen kinematik und mechanik. *Zeitschrift für Physik*, 43(3-4):172–198, 1927.
- [20] Geoffrey Hinton. The forward-forward algorithm: Some preliminary investigations. *arXiv preprint arXiv:2212.13345*, 2022.
- [21] Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- [22] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- [23] Sabine Kastner and Leslie G Ungerleider. Mechanisms of visual attention in the human cortex. *Annual review of neuroscience*, 23(1):315–341, 2000.
- [24] Seyed Mehran Kazemi, Rishab Goel, Sepehr Eghbali, Janahan Ramanan, Jaspreet Sahota, Sanjay Thakur, Stella Wu, Cathal Smyth, Pascal Poupart, and Marcus Brubaker. Time2vec: Learning a vector representation of time. *arXiv preprint arXiv:1907.05321*, 2019.
- [25] Bernard O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931.
- [26] Yihao Liu et al. Mono-forward: Backpropagation-free algorithm for efficient neural network training harnessing local errors. *arXiv preprint arXiv:2501.09238*, 2025.
- [27] Wayne M. McDonnell. Analysis and assessment of the Gateway process. <https://www.cia.gov/readingroom/docs/CIA-RDP96-00788R001700210016-5.pdf>, 1983. U.S. Army Intelligence and Security Command (INSCOM); declassified, CIA Reading Room record CIA-RDP96-00788R001700210016-5.
- [28] Evan Miller. Attention is off by one. <https://www.evanmiller.org/attention-is-off-by-one.html>, 2023.
- [29] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023.
- [30] OilProducts. Abstractinator. <https://github.com/OilProducts/abstractinator>, 2025. Byte-latent encoder with a multi-stage residual vector-quantization bottleneck.
- [31] Artidoro Pagnoni et al. Byte latent transformer: Patches scale better than tokens. *arXiv preprint arXiv:2412.09871*, 2024.

- [32] Raphael Pisoni. Soap bubbles and attention sinks: The theory and history of the HALO-loss. <https://pisoni.ai/posts/halo/>, 2026. HALO = Hyperspherical Alignment & Latent Optimization.
- [33] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. In *ICLR MATH-AI Workshop*, 2022.
- [34] Chenze Shao et al. Continuous autoregressive language models. *arXiv preprint arXiv:2510.27688*, 2025.
- [35] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.
- [36] Hava T. Siegelmann and Eduardo D. Sontag. On the computational power of neural nets. *Journal of Computer and System Sciences*, 50(1):132–150, 1995.
- [37] Yixuan Su, Tian Lan, Yan Wang, Dani Yogatama, Lingpeng Kong, and Nigel Collier. A contrastive framework for neural text generation. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [38] Floris Takens. Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence, Warwick 1980*, pages 366–381. Springer, 1981.
- [39] Matthew Tancik, Pratul Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- [40] Matus Telgarsky. Benefits of depth in neural networks. In *Conference on Learning Theory (COLT)*, pages 1517–1539, 2016.
- [41] Jayden Teoh, Manan Tomar, Kwangjun Ahn, Edward S. Hu, Tim Pearce, Pratyusha Sharma, Akshay Krishnamurthy, Riashat Islam, Alex Lamb, and John Langford. Next-latent prediction transformers learn compact world models, 2025.
- [42] Stefan Treue and John HR Maunsell. Attentional modulation of visual motion processing in cortical areas mt and mst. *Nature*, 382(6591):539–541, 1996.
- [43] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in neural information processing systems*, volume 30, 2017.
- [44] Hermann Weyl. Über die gleichverteilung von zahlen mod. eins. *Mathematische Annalen*, 77(3):313–352, 1916.